

Prediction of Outcome from Spatial Protein Profiling of Triple-Negative Breast Cancers

Corresponding Author: Dr Jeffrey Chuang

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This manuscript proposed a method called SparTile that analyzes spatial proteomic data without requiring cell segmentations. The authors used this method to identify TME markers including Ki67+ and Vimentin+ which correlate with negative outcomes. The author further defined distance metrics that correlate with patient survival.

The overall method seems straightforward: tile + clustering for the discovery of new biomarkers. Based on the result, the tool is effective. However, confirmation is required to evaluate its performance against cell segmentation approaches and its robustness.

Comments:

1. The author mentioned that cell segmentation methods are limited. It would be good to compare segmentation-based approaches and determine if the clusters identified with SparTile can also be discovered using these methods.
2. The p-value seems to be on the edge of the significance threshold. One reason could be that the granularity level is not correct. This can be adjusted by defining the number of clusters chosen. It may be worth to try different numbers of clusters (e.g. from 5 to 50) and see if:
 - a. more refined clusters (better correlation with survival) are obtainable.
 - b. the results are still robust.
3. The method described here seems very lightweight to implement, I encourage the authors to make SparTile available as a package on Pypi to make this tool more accessible.

Reviewer #2

(Remarks to the Author)

SparTile is a new computational method designed to analyze complex tumor environments in cancer tissue without needing to separate or label individual cells. By looking at the spatial relationships between different proteins in images of tumor samples, SparTile can identify distinct microenvironments (MEs) that are linked to how aggressive the cancer is and how likely a patient is to survive. In triple-negative breast cancer, SparTile found specific patterns like areas rich in certain proteins (e.g., Ki67, vimentin, MX1, or AXL) that are associated with poor outcomes, while others with immune cells (like CD8+ T cells) suggest better survival. It also found that the physical distance between tumor and immune cells, especially myeloid cells, can strongly predict survival chances.

Comments:

1. How were the 88 tumors on the TMA selected? Were they all from the same tumor grade, stage? Was the entire tissue issue on the TMA core used for the analysis or was it annotated by a pathologist to avoid fibrotic/ necrotic areas? More details are needed on that in the methods section.
2. IMC can be used in 1 tissue for the entire 47 marker panel. Why was it divided between two consecutive tissues? Was the rational to avoid signal spillover between the channels? If so, there are pipeline already developed like CATALYST by

Bodenmiller lab at the university of Zurich to correct for that! <https://pubmed.ncbi.nlm.nih.gov/29605184/> A thorough explanation is needed especially that the two panels used between the 2 slides were different without shared markers meaning that the two slides could offer distinctively different cell layers.

3. What are the size of the tiles and roughly how many cells do you expect them to have? A thorough explanation is needed for the annotation of these tiles and the limitations of this way of doing tile-level annotation instead of single-cell level.

4. In the text, it is mentioned that one of the slides is stained with general markers to identify the tissue regions whereas the other slide is stained with immune markers, but by looking at the table in figure 1A, and I see 1 marker under "immune" and this can be confusing. So, I recommend naming CD45 a different name. or consider combining it with other immune markers for consistency.

5. Fig1C —> from my understanding the unbiased ME identification is followed by a more granular characterization of clusters based on the protein expression data. However, the figure suggests that they are two independent steps. Please comment here.

6. Add color map to figure 2D to be able to map tumor or stromal regions to the annotated tissue map with the tiles

7. Figure 3 —> it is easier to have a letter for each individual figure instead of referring to figures by Fig.3A top and Fig.3A middle, etc. And same problem in Fig3 bottom to add the color panels for the tiled regions otherwise they have no meaning comparing them to the raw IMC data. Also the color panels used for figure 3 and figure 2 is different from what was used in figure 1D so it is absolutely critical to make sure you are consistent across figures so the reader can always compare the figures against each other.

8. While the authors highlight the difficulty of segmentation level analysis, there is still huge benefit in segmenting single cells and getting the single-cell protein expression data. It is insightful to annotate cell types more confidently instead of annotating an entire tile to a tissue region with it having other cell types. More discussion is needed on the limitation on this segmentation-free approach.

9. In the virtual H&E generation: why was CD3 and CD30 used as part of the markers to identify the nucleus even though they are membrane markers? Isn't DNA1 and DNA3 sufficient for this task? If not, I recommend adding a histone marker for this task.

10. Also, why markers like collagen and αSMA were used to identify the cytoplasm when they can mostly be secreted in the extracellular matrix and not only specific for the cytoplasm? Was pan cytokeratin wasn't used for the cytoplasm detection?

11. Add the raw images of the IMC for all markers (47) as well the virtual H&Es in the supplementary. It is critical for the reader to see the data and judge the staining quality.

12. Also please in the methods section the details about which antibodies were conjugated to which metal tags and the rationale behind these choices.

13. If performing an additional experiment to combine the entire 47 markers on one tissue is not possible or feasible then I would like to see some validation on the one slide where the single-cell segmentation is compared to tile-level annotation. For example, the most significant finding of this paper was the importance of the spatial distance between myeloid cells and tumor cells, so having several samples stained with a panel consisting of some myeloid cell markers + tumor markers and then perform the same analysis side by side on the cell-level and the tile-level and see if the observations will still be there or maybe stronger?

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The author has addressed all my concerns. I am in support of publication of this work.

Here are my additional comments about feedback from reviewer 2. It would only require minor updates to the text. Otherwise, I think the authors have addressed most of the points. I support the publication of this manuscript.

2. The splitting channels into two and performing coregistration is rather distracting. I suggest summarizing the process with a few sentences and moving the overly technical details to supplementary and mentioning it as a limitation of the data acquisition method.

8. I would add that the segmentation-free and segmentation-based approaches are not mutually exclusive, and can be combined to provide more robust analysis.

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Reviewer response of manuscript:

Prediction of Outcome from Spatial Protein Profiling of Triple-Negative Breast Cancers

Ali Foroughi pour¹, Te-Chia Wu¹, Javad Noorbakhsh¹, Jan Martinek¹, Santhosh Sivajothi¹, Paul Robson^{1,2}, Karolina Palucka^{1,2**}, Jeffrey H. Chuang^{1,2**}

1-The Jackson Laboratory for Genomic Medicine, Farmington, CT, USA. 2-UCONN Health, Department of Genetics and Genome Sciences, Farmington, CT, USA. **co-corresponding authors

We thank the reviewers for the constructive feedback. We have edited the manuscript based on reviewer comments, which are highlighted in red in the manuscript. Below we have provided point-by-point responses to each comment.

Reviewer #1 (Remarks to the Author):

This manuscript proposed a method called SparTile that analyzes spatial proteomic data without requiring cell segmentations. The authors used this method to identify TME markers including Ki67+ and Vimentin+ which correlate with negative outcomes. The author further defined distance metrics that correlate with patient survival.

The overall method seems straightforward: tile + clustering for the discovery of new biomarkers. Based on the result, the tool is effective. However, confirmation is required to evaluate its performance against cell segmentation approaches and its robustness.

Comments:

1. The author mentioned that cell segmentation methods are limited. It would be good to compare segmentation-based approaches and determine if the clusters identified with SparTile can also be discovered using these methods.

Response: We agree that comparison with segmentation-based methods can clarify the value of SparTile. Therefore, we have added a section “Comparison with segmentation-based analysis”. We find that SparTile can distinguish MEs more consistently across hyperparameters than segmentation-based approaches. In addition, the SparTile spatial distance metric yielded more consistent associations with survival than those based on cellular distances. We have edited the main text to describe these points:

“Comparison with segmentation-based analysis

To further assess the robustness and interpretability of SparTile we compared it to cell segmentation-based analysis. For segmentation and classification, we analyzed only the tissue slice corresponding to panel 1. This is because individual cells are sometimes not large enough to span the adjacent slices (Supplementary Figure 1), and panel 2 lacks markers for diverse cell phenotyping on its own (methods, Supplementary File 2). We used Mesmer (Greenwald et. al, 2022) for cell segmentation and hierarchical clustering to identify cell phenotypes (methods,

Supplementary File 2, Supplementary Fig. 6). We then defined cell segmentation-based microenvironments based on the cell types around each individual cell. For each center cell, the prevalence of cell types within radius $R=30$ was calculated. Hierarchical clustering of these prevalence vectors was then applied, yielding $N=25$ MEs (methods, Supplementary Fig. 6).

For a matched comparison to the cell segmentation analysis, we re-applied SparTile to markers of only panel 1 (MEs as a function of hyperparameters are provided in Supplementary File 1). Most MEs correlating with survival in the two-panel analysis were re-identified. For instance, the MX1+ tumor myeloid ME, Ki67+ ME, vimentin+ tumor ME, T-cell MEs, and B-cell MEs were re-identified using panel 1 alone. This is not surprising as these MEs are described by markers present in panel 1. Furthermore, the SparTile distance between tumor and myeloid markers (Fig. 4) robustly separated survival curves using the panel 1 data alone (Supplementary Fig. 7a, log-rank p -value <0.01).

Segmentation-based analysis consistently identified several MEs also found by SparTile, such as the tumor ME with epithelial to mesenchymal transition (cluster 7), Ki67+ tumor MEs (clusters 2, 6,15, and 19) and the B-cell dominated ME (clusters 9 and 11) (Supplementary Fig. 6). The segmentation-based tumor ME with epithelial to mesenchymal transition (cluster 7, $T=0.01$, log rank p -value $=0.0009$) and CD8+ T-cell ME (cluster 14, $T=0.01$, log rank p -value $=0.05$) each showed strong correlation with survival risk, but contrary to expectations the Ki67+ tumor MEs showed weak survival associations (clusters 2, 6,15,and 19, $T=0.01$, log rank p -value $=0.25$) (Supplementary Fig. 7). In contrast, with SparTile the p -value is 0.052. Moreover, the MX1+ tumor-myeloid signal found by SparTile was split among several MEs when analyzed by segmentation, regardless of hyperparameters, hindering identification of the survival risk for this ME (methods, Supplementary Fig. 6, Supplementary File 2).

We next tested if a segmentation-based spatial distance between tumor and myeloid cells could separate survival curves (methods). We measured the average distance from each tumor cell to the 5 nearest myeloid cells (denoted by $T \rightarrow M$, methods) and vice versa (denoted by $M \rightarrow T$). We additionally computed the average of both distances (denoted by $M \leftrightarrow T$). We only observed a significant association between distance and improved survival in the $M \rightarrow T$ case (Supplementary Fig. 8). None of these metrics were significantly associated with survival in a multivariate Cox model (Supplementary Figure 8), although $M \rightarrow T$ correlated with the SparTile distance (Pearson Correlation coefficient $=0.54$, p -value $=7e-8$). We externally tested our segmentation-based distances on the Basel (Jackson et al, 2020) and METABRIC IMC (Ali et al, 2020) datasets (Supplementary Fig. 8). We observed a lack of consistency across the 3 datasets. For example, for the $M \rightarrow T$ metric, in our JAX dataset and the Basel datasets, smaller $M \rightarrow T$ correlated with higher survival risk, while in METABRIC it correlated with lower survival risk, though weakly (Supplementary Fig. 8). For the $T \rightarrow M$ metric, the direction of effect was also inconsistent among the three datasets. This was the case for the $T \leftrightarrow M$ metric as well. Moreover, even when the effects were in the expected direction (higher tumor-myeloid distance associated with better survival), p -values were often weak. In only the $T \rightarrow M$ JAX and $M \leftrightarrow T$ METABRIC cases were the p -values <0.01 . No cases showed statistically significant effects in multivariate Cox models (Supplementary Fig. 8)."

We have also edited the discussion to summarize and explain these findings:

“Segmentation-based methods are valuable but have challenges, including error-prone cell segmentation, signal spillage from adjacent cells and MEs, and the computational burden of identifying and representing cellular neighborhood (Windhager et al., 2023 and Milosevic, 2023). Recent cell phenotyping tools aim to correct for signal spillage (Milosevic, 2023). This improves cell phenotyping but may filter out information important to ME inference. Graph neural networks have emerged as potential tools for understanding microenvironments from segmented cells (Siao et al, 2024) but their interpretability is limited, and incorporating structural and signaling proteins into their node representations remains complex...We also note that SparTile identifies MEs in an unbiased fashion and encodes relative spatial distribution of arbitrary sets of protein markers. As the number of combinations to test for spatial distribution analysis are computationally intractable, SparTile’s ME identification step can be used to narrow down the search. Another strength of SparTile is that MEs are more directly computed from pixel data, while segmentation-based models have multiple steps involving complex biological assumptions, including but not limited to, cell segmentation, cell phenotyping, and ME identification. This likely explains the lower robustness of segmentation approaches in identifying survival associations. Nonetheless, cell-type information remains essential for many downstream applications, such as designing experiments using organoid models or patient-derived xenografts for targeted therapies. Although we have shown that cellular and spatial composition of SparTile’s MEs can be estimated post hoc, segmentation-based methods provide an essential complementary approach for clarifying cell-level interpretability.”

2. The p-value seems to be on the edge of the significance threshold. One reason could be that the granularity level is not correct. This can be adjusted by defining the number of clusters chosen. It may be worth to try different numbers of clusters (e.g. from 5 to 50) and see if:

a. more refined clusters (better correlation with survival) are obtainable.

b. the results are still robust.

Response: We agree with the reviewer that hyperparameter optimization is essential to obtain optimal results. As our study was aimed at discovery, in the original submission hyperparameters were selected based on prioritizing biological interpretability. To address the reviewer’s concern, we have reported the results across multiple hyperparameter values. In general, we find that the SparTile MEs described in the original manuscript are robust across a range of hyperparameters. We have clarified our standard choices: “SparTile uses sparse non-negative matrix factorization (sNMF) to learn low-dimensional representations of tiles of size $S \times S$ microns within each region type (methods, Fig. 2A). Using these representations, we then learned $D=20$ clusters in the tumor ($S=30$), $D=20$ clusters in the stroma ($S=30$), and $D=10$ clusters in the immune ($S=20$) region type (methods, Figs. 1D, 2B, and 2C).” Throughout the text, we now further describe robustness of results to hyperparameters.

For example, for the MX1+ tumor myeloid ME:

“This ME was robustly identified and separated survival curves under other hyperparameters (e.g. $D=15$, $S=40$, $T=0.01$, log-rank p-value=0.015, $D=20$ and $S=20$, log rank p-value=0.05, Supplementary Fig. 2, methods).”

For t4, a vimentin+ tumor ME:

“This ME was robustly identified and separated survival curves under other hyperparameters (e.g. D=20 and S=20 log-rank p-value=4e-4, D=15 and S=40, log rank p-value=2e-5, Supplementary Fig. 2).”

For KI67+ MEs:

“This ME was robustly identified and separated survival curves under other hyperparameters (e.g. D=15 and S=20 for both tumor and stroma region types, T=0.01 log-rank p-value=0.06, D=20 and S=40 for both tumor and stroma region types, T=0.01 log rank p-value=0.02, Supplementary Fig. 2).”

Additional cases are described in the text. See revised manuscript for details.

We have also included the top contributors to each NMF component for a wider range of hyperparameters in Supplementary File 1, to quantify how each ME is robustly identified. In addition, we have edited the methods regarding how the default hyperparameters were chosen:

“We observed that using a large number of dimensions D (e.g. D=50) reduces sNMF stability and results in noise determining many sNMF clusters (Supplementary File 1). Therefore we used S=30 and D=20 for tumor and stroma regions, and used S=20 and D=10 for immune regions.”

3. The method described here seems very lightweight to implement, I encourage the authors to make SparTile available as a package on Pypi to make this tool more accessible.

Response: As recommended by the reviewer we have uploaded our code, a brief documentation, and a jupyter notebook example going over the SparTile pipeline accessible at: <https://github.com/aforoughipour/SparTile>. The Github link is also provided in the main manuscript:

“The code associated with the manuscript and a minimal working Jupyter notebook example are available on GitHub (<https://github.com/aforoughipour/SparTile>).”

Reviewer #2 (Remarks to the Author):

SparTile is a new computational method designed to analyze complex tumor environments in cancer tissue without needing to separate or label individual cells. By looking at the spatial relationships between different proteins in images of tumor samples, SparTile can identify distinct microenvironments (MEs) that are linked to how aggressive the cancer is and how likely a patient is to survive. In triple-negative breast cancer, SparTile found specific patterns like areas rich in certain proteins (e.g., Ki67, vimentin, MX1, or AXL) that are associated with poor outcomes, while others with immune cells (like CD8+ T cells) suggest better survival. It also found that the physical distance between tumor and immune cells, especially myeloid cells, can strongly predict survival chances.

Comments:

1. How were the 88 tumors on the TMA selected? Were they all from the same tumor grade, stage? Was the entire tissue on the TMA core used for the analysis or was it annotated by a pathologist to avoid fibrotic/ necrotic areas? More details are needed on that in the methods section.

Response: We thank the reviewer for this comment. We have included additional information in the methods to explain the process used for TMA generation:

“Tissue Microarrays (TMAs) were obtained from Yale School of Medicine (IRB#2019-088). A search for TNBC was done from 2000-2012. All cases that were available in the archives were collected. The grade and stage information were not used as an inclusion criteria. Expert review identified the block with the most tumor present and annotated a representative region on an H&E slide from that block. The final core was extracted from the pathologist annotated region. TMA spots were 600 to 800 microns in diameter. Two adjacent 5-micron cuts were obtained for each spot. IMC was performed on TMA spots. Images corresponding to cell lines, relapsing or metastatic patients, or patients who received neoadjuvant therapy were removed from downstream analysis. 5 patients had two microarray spots, while all other patients had 1 spot. The images corresponding to 88 primary TNBC patients were used for downstream analysis. These patients were monitored, received adjuvant chemotherapy, adjuvant radiotherapy, or both adjuvant chemotherapy and radiotherapy.”

2. IMC can be used in 1 tissue for the entire 47 marker panel. Why was it divided between two consecutive tissues? Was the rationale to avoid signal spillover between the channels? If so, there are pipeline already developed like CATALYST by Bodenmiller lab at the university of Zurich to correct for that! <https://pubmed.ncbi.nlm.nih.gov/29605184/> A thorough explanation is needed especially that the two panels used between the 2 slides were different without shared markers meaning that the two slides could offer distinctively different cell layers.

Response: We agree it would have been preferable to use a single panel to profile all tissue samples. However, it was not feasible for us to include all 47 unique markers in a single panel due to limitations in the availability of metal isotopes for conjugation. While some publications at the time had demonstrated the use of alternate isotopes in cytometry applications, they had not been rigorously tested for signal spillover or stability. To ensure data quality and reproducibility, we therefore limited our selection to isotopes provided by Standard BioTools, which were available as validated conjugation kits. The following has been added to the methods to clarify this point:

“For IMC panel design, where available, pre-conjugated antibodies were obtained from Standard BioTools. In Panel 1, we utilized all available isotopes except for 172Yb, assigning each to a distinct marker. To incorporate additional markers—particularly those focused on myeloid cell populations—we designed Panel 2, which includes 11 unique markers not present in Panel 1. While Panel 2 shares 20 markers with Panel 1 to ensure continuity and comparability, the 11 additional markers could not be accommodated in Panel 1 due to the unavailability of 11 distinct isotopes. Therefore, the two-panel approach was a necessary compromise to achieve broad cellular and phenotypic coverage while working within the technical constraints of the IMC platform. For custom-conjugated

antibodies, isotope selection followed a set of strategic guidelines to optimize signal quality and minimize spillover: (a) Markers with high expression were assigned to isotopes known to produce minimal spillover. (b) Markers with low expression were paired with isotopes that generate strong signal intensity. (c) Markers with low expression were also prioritized for isotopes that are less susceptible to receiving spillover. (d) When necessary, low-expression markers were assigned to isotopes that may produce higher spillover, balancing overall panel performance.”

3. What are the size of the tiles and roughly how many cells do you expect them to have? A thorough explanation is needed for the annotation of these tiles and the limitations of this way of doing tile-level annotation instead of single-cell level.

Response: We have added Supplementary Figure 11 where the distribution of number of cells per tile is reported across different tile sizes and tissue regions. For the question of tile-based vs. cell-based analysis, we have improved the discussion section:

“Segmentation-based methods are valuable but have challenges, including error-prone cell segmentation, signal spillage from adjacent cells and MEs, and the computational burden of identifying and representing cellular neighborhood (Windhager et al., 2023 and Milosevic, 2023). Recent cell phenotyping tools aim to correct for signal spillage (Milosevic, 2023). This improves cell phenotyping but may filter out information important to ME inference. Graph neural networks have emerged as potential tools for understanding microenvironments from segmented cells (Siao et al, 2024) but their interpretability is limited, and incorporating structural and signaling proteins into their node representations remains complex. Additionally, the influence of network architecture and hyperparameter choices on the resulting representations is not fully understood.

SparTile addresses these limitations by employing a tile-based approach that encodes spatial relationships and facilitates the direct identification of distinct microenvironments. Compared to single-pixel analyses (e.g. Liu et al, 2023), tiles provide richer contextual information, reducing the need for extensive preprocessing and hyperparameter tuning. SparTile has conceptual advantages over cell-cell contact-based metrics (e.g. Wang et al 2023), which fail to account for dynamic cellular states, such as the persistence of T-cell activation post-interaction with cancer cells. Because SparTile integrates such dynamics, it provides a more complete characterization of spatial information.

We also note that SparTile identifies MEs in an unbiased fashion and encodes relative spatial distribution of arbitrary sets of protein markers. As the number of combinations to test for spatial distribution analysis are computationally intractable, SparTile’s ME identification step can be used to narrow down the search. Another strength of SparTile is that MEs are more directly computed from pixel data, while segmentation-based models have multiple steps involving complex biological assumptions, including but not limited to, cell segmentation, cell phenotyping, and ME identification. This likely explains the lower robustness of segmentation approaches in identifying survival associations. Nonetheless, cell-type information remains essential for many downstream applications, such as designing experiments using organoid models or patient-derived xenografts for targeted

therapies. Although we have shown that cellular and spatial composition of SparTile's MEs can be estimated post hoc, segmentation-based methods provide an essential complementary approach for clarifying cell-level interpretability."

We note that Reviewer 1 asked for a comparison of MEs identified by SparTile vs. MEs identified by cell segmentation-based approaches, and the revised Results section now includes a new subsection on this topic. These new analyses indicate that SparTile MEs are more robust than segmentation-based MEs, as alluded to in the revised discussion text. For details, see response to comment 1 from Reviewer 1.

4. In the text, it is mentioned that one of the slides is stained with general markers to identify the tissue regions whereas the other slide is stained with immune markers, but by looking at the table in figure 1A, and I see 1 marker under "immune" and this can be confusing. So, I recommend naming CD45 a different name. or consider combining it with other immune markers for consistency.

Response: As suggested, we have renamed the "Immune" column to "Gen. Immune" to emphasize that CD45 is a broad-based immune cell marker, while other columns determine distinct immune cell types.

5. Fig1C —> from my understanding the unbiased ME identification is followed by a more granular characterization of clusters based on the protein expression data. However, the figure suggests that they are two independent steps. Please comment here.

Response: We apologize for the confusion. As Figure 1C suggests, unbiased ME identification is separate from the spatial characterization of protein markers (this relates to the Results section "Spatial distribution of tumors and myeloid cell markers is prognostic of survival"). In principle, one could construct the spatial distance for any two groups of protein markers; however, this problem is computationally challenging due to the large number of potential combinations. Therefore, the results of the ME identification step were used to narrow down the search space. We have edited the discussion section to reflect this point:

"We also note that SparTile identifies MEs in an unbiased fashion and encodes relative spatial distribution of arbitrary sets of protein markers. As the number of combinations to test for spatial distribution analysis are computationally intractable, SparTile's ME identification step can be used to narrow down the search."

6. Add color map to figure 2D to be able to map tumor or stromal regions to the annotated tissue map with the tiles

Response: We have added color maps to Figure 2D.

7. Figure 3 —> it is easier to have a letter for each individual figure instead of referring to figures by Fig.3A top and Fig.3A middle, etc. And same problem in Fig3 bottom to add the color panels for the tiled regions otherwise they have no meaning comparing them to the raw IMC data. Also the color panels used for figure 3 and figure 2 is different from what was used in figure 1D so it is absolutely critical to make sure you are consistent across figures so the reader can always compare the figures against each other.

Response: We thank the Reviewer for finding these improvements. We have edited Figure 3 as suggested.

8. While the authors highlight the difficulty of segmentation level analysis, there is still huge benefit in segmenting single cells and getting the single-cell protein expression data. It is insightful to annotate cell types more confidently instead of annotating an entire tile to a tissue region with it having other cell types. More discussion is needed on the limitation on this segmentation-free approach.

Response: We thank the reviewer for this comment. We certainly agree that the understanding of single cell protein expression is fundamental. The revised discussion indicates how segmentation-based methods and SparTile are complementary:

“Nonetheless, cell-type information remains essential for many downstream applications, such as designing experiments using organoid models or patient-derived xenografts for targeted therapies. Although we have shown that cellular and spatial composition of SparTile’s MEs can be estimated post hoc, segmentation-based methods provide an essential complementary approach for clarifying cell-level interpretability.”

9. In the virtual H&E generation: why was CD3 and CD30 used as part of the markers to identify the nucleus even though they are membrane markers? Isn’t DNA1 and DNA3 sufficient for this task? If not, I recommend adding a histone marker for this task.

Response: We thank the reviewer for this comment as our initial explanation was insufficient. Indeed CD3 and CD20 markers are membrane markers, but as cells are 3D objects and lymphocytes tend to be small in size, we observed non-zero expression of CD3 and CD20 on or immediately adjacent to pixels that stained positive for DNA. As the reviewer notes, these signals are due to membrane directly above or below the nucleus. However, we observed that adding them as nuclear markers resulted in darker lymphocytes in the virtual H&E images, more similar to real H&Es. This effect was reviewed by a clinical pathologist in our lab. We have edited the methods section to reflect this:

“While the input channels are called nucleic or cytoplasmic in FalseColor (Serafin et al, 2020), the goal is to mimic hematoxylin and eosin staining. We observed that adding protein markers found in both the cytoplasm and extracellular matrix improves the virtual H&E images. Furthermore, we observed non-zero expression of membrane proteins CD3 and CD20 on or immediately adjacent to pixels in the 2D space that stained positive for DNA. This signal arises from membrane signal directly above or below the nucleus in the z-axis and is notable especially for small cells such as lymphocytes. Due to this effect, we found that treating CD3 and CD20 as nuclear markers generates darker lymphocyte nuclei in the virtual H&Es and made them appear more similar to real H&E images (Supplementary Fig. 10).”

10. Also, why markers like collagen and aSMA were used to identify the cytoplasm when they can mostly be secreted in the extracellular matrix and not only specific for the cytoplasm? Was pan cytokeratin wasn’t used for the cytoplasm detection?

Response: We thank the reviewer for this comment as our initial explanation was insufficient. We observed that adding cytoplasmic protein markers that are also expressed in the extra-cellular matrix helps generate more realistic H&E images. This effect was reviewed by a clinical pathologist in our lab. We did not use Pan_CK because it was only present in 1 panel, and as the goal was to register the 2 panels we only used shared markers.

11. Add the raw images of the IMC for all markers (47) as well the virtual H&Es in the supplementary. It is critical for the reader to see the data and judge the staining quality.

Response: We have uploaded png images of all 47 markers as well as virtual H&E images on Zenodo (available at <https://zenodo.org/records/17056591>).

12. Also please in the methods section the details about which antibodies were conjugated to which metal tags and the rationale behind these choices.

Response: The clones, vendors, and the metal tags used are provided in Supplementary File 2 (sheet 2), and we have edited the methods section to describe the rationale behind panel design:

“For IMC panel design, where available, pre-conjugated antibodies were obtained from Standard BioTools. In Panel 1, we utilized all available isotopes except for ¹⁷²Yb, assigning each to a distinct marker. To incorporate additional markers—particularly those focused on myeloid cell populations—we designed Panel 2, which includes 11 unique markers not present in Panel 1. While Panel 2 shares 20 markers with Panel 1 to ensure continuity and comparability, the 11 additional markers could not be accommodated in Panel 1 due to the unavailability of 11 distinct isotopes. Therefore, the two-panel approach was a necessary compromise to achieve broad cellular and phenotypic coverage while working within the technical constraints of the IMC platform. For custom-conjugated antibodies, isotope selection followed a set of strategic guidelines to optimize signal quality and minimize spillover: (a) Markers with high expression were assigned to isotopes known to produce minimal spillover. (b) Markers with low expression were paired with isotopes that generate strong signal intensity. (c) Markers with low expression were also prioritized for isotopes that are less susceptible to receiving spillover. (d) When necessary, low-expression markers were assigned to isotopes that may produce higher spillover, balancing overall panel performance.”

13. If performing an additional experiment to combine the entire 47 markers on one tissue is not possible or feasible then I would like to see some validation on the one slide where the single-cell segmentation is compared to tile-level annotation. For example, the most significant finding of this paper was the importance of the spatial distance between myeloid cells and tumor cells, so having several samples stained with a panel consisting of some myeloid cell markers + tumor markers and then perform the same analysis side by side on the cell-level and the tile-level and see if the observations will still be there or maybe stronger?

Response: Due to the limitations discussed above, it was not possible for us to do all 47 markers on one tissue. However, Panel 1 contains both tumor and myeloid markers, so we performed the suggested segmentation-based analysis on panel 1. This analysis is discussed in detail in the

response to Reviewer 1, comment 1. These new analyses indicate that SparTile yields more robust associations with survival compared to segmentation-based approaches.

Reviewer response of manuscript:

Prediction of Outcome from Spatial Protein Profiling of Triple-Negative Breast Cancers

We thank the reviewers for their valuable feedback. It has helped the authors improve clarity of the manuscript. Here we provide authors' point-by-point response to reviewer comments:

Comment 1: The splitting channels into two and performing coregistration is rather distracting. I suggest summarizing the process with a few sentences and moving the overly technical details to supplementary and mentioning it as a limitation of the data acquisition method.

Response: The journal requirement for the methods section is that "All descriptions of Methods should be provided in the main manuscript file and not in the Supplementary Information." Therefore, the authors cannot move this step to the supplementary. However, per reviewer's suggestion, we have edited the registration section to make it more concise and clearer:

"The similarity transform of the Airlab package (v0.2.1, Sandkühler et al, 2018) was used to register grayscale virtual H&E images to each other. The second panel was transformed to match the coordinates of the first panel. Max projection was used for IMC markers common between the two panels."

Comment 2: I would add that the segmentation-free and segmentation-based approaches are not mutually exclusive, and can be combined to provide more robust analysis.

Response: We agree with the reviewer that segmentation-based methods and SparTile are not mutually exclusive but are complementary. As this point was not made clear in the previously revised discussion, we have edited the discussion to clarify this point:

"...cell-type information remains essential for many downstream applications, such as designing experiments using organoid models or patient-derived xenografts for targeted therapies. Therefore, segmentation-based methods and SparTile are not mutually exclusive, but complementary. Although we have shown that cellular and spatial composition of SparTile's MEs can be estimated post hoc, segmentation-based methods provide an essential complementary approach for clarifying cell-level interpretability."